CO1-ASSESSMENT TOOL-3

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COURSE: Deep Learning

COURSE CODE: MLAO406

Experiment 1:

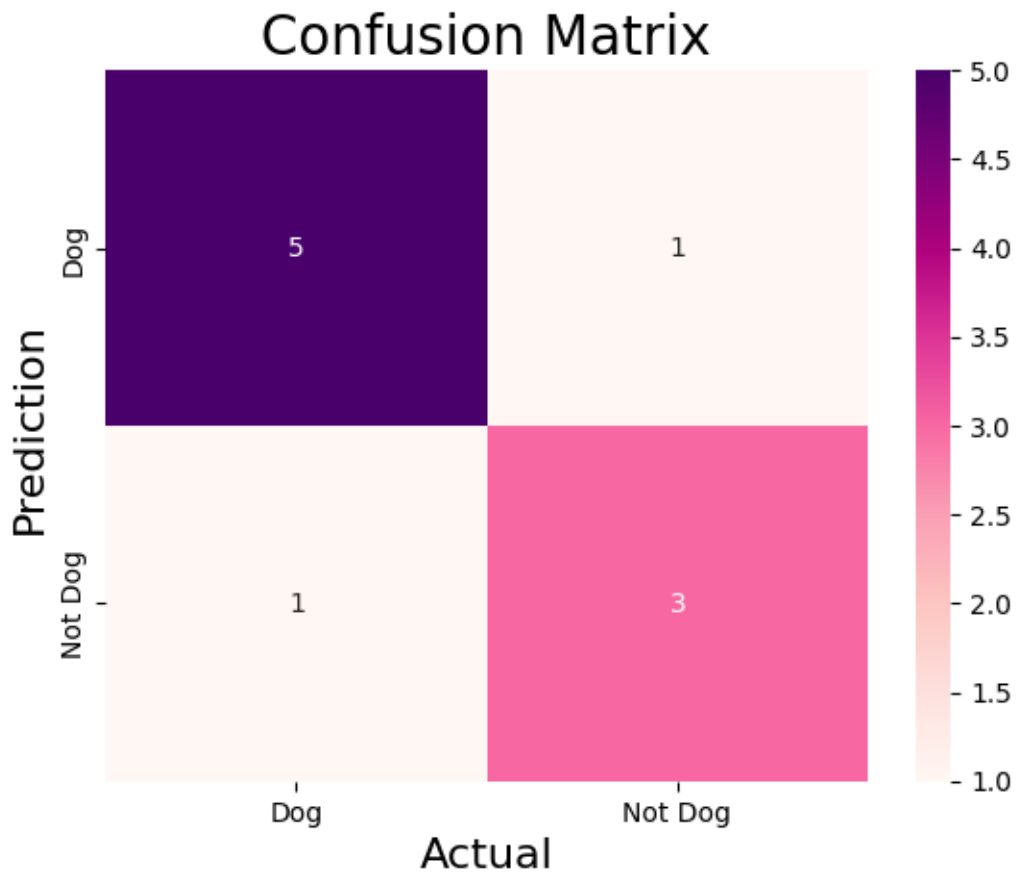
Aim: To demonstrate confusion matrix using python

Program:

```
import numpy as np
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
actual = np.array(['Dog', 'Dog', 'Dog', 'Not Dog', 'Dog', 'Not Dog', 'Dog', 'Dog', 'Not Dog', 'Not Dog'])
predicted = np.array(['Dog', 'Not Dog', 'Dog', 'Not Dog', 'Dog', 'Dog', 'Dog', 'Dog', 'Not Dog', 'Not Dog'])
# Explicitly define labels to ensure correct alignment in the confusion matrix
labels = ['Dog', 'Not Dog']
conf_matrix = confusion_matrix(actual, predicted, labels=labels)
sns.heatmap(
  conf_matrix,
  annot=True,
  fmt='g',
  xticklabels=labels,
```

```
    yticklabels=labels,
    cmap='RdPu'
)
plt.ylabel("Prediction", fontsize=16)
plt.xlabel("Actual", fontsize=16)
plt.title("Confusion Matrix", fontsize=20)
plt.show()
```

OUTPUT:



Experiment 2:

Aim: To demonstrate 2 class confusion matrix using python

Program:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import (
  accuracy_score,
  f1_score,
  precision_score,
  recall_score,
  classification_report,
  confusion_matrix
)
wine = load_wine()
data = pd.DataFrame(data=wine.data, columns=wine.feature_names)
data['Target'] = wine.target
data = data[data['Target'] != 2]
x = data.drop('Target', axis=1)
y = data['Target']
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3, random_state=1)
model = DecisionTreeClassifier(random_state=1)
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
accuracy = accuracy_score(y_test, y_pred)
```

```
print("accuracy:", accuracy)
class_report = classification_report(y_test, y_pred, target_names=wine.target_names[:2])
print("Classification Report:\n", class_report)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1 Score: {f1:.2f}")
conf_matrix = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(
   conf_matrix,
   annot=True,
   fmt='d',
   cmap='PuBuGn',
   xticklabels=wine.target_names[:2],
   yticklabels=wine.target_names[:2]
)
plt.xlabel("predicted label")
plt.ylabel("true label")
plt.title("confusion matrix")
plt.show()
```

OUTPUT:

accuracy: 0.9743589743589743

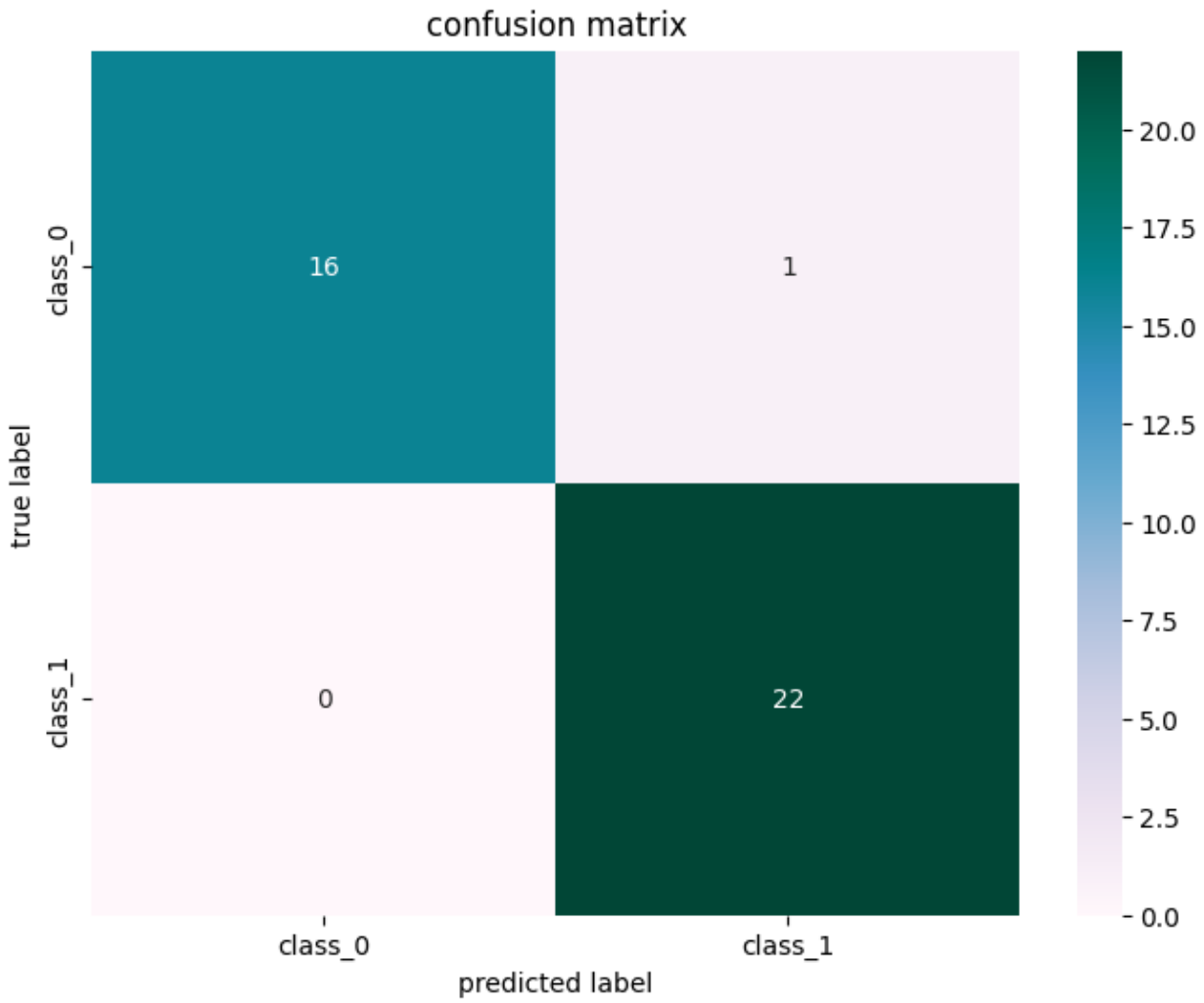
Classification Report:

	precision	recall	f1-score	support
class_0	1.00	0.94	0.97	17
class_1	0.96	1.00	0.98	22
accuracy			0.97	39
macro avg	0.98	0.97	0.97	39
weighted avg	0.98	0.97	0.97	39

Precision: 0.96

Recall: 1.00

F1 Score: 0.98



Experiment 3:

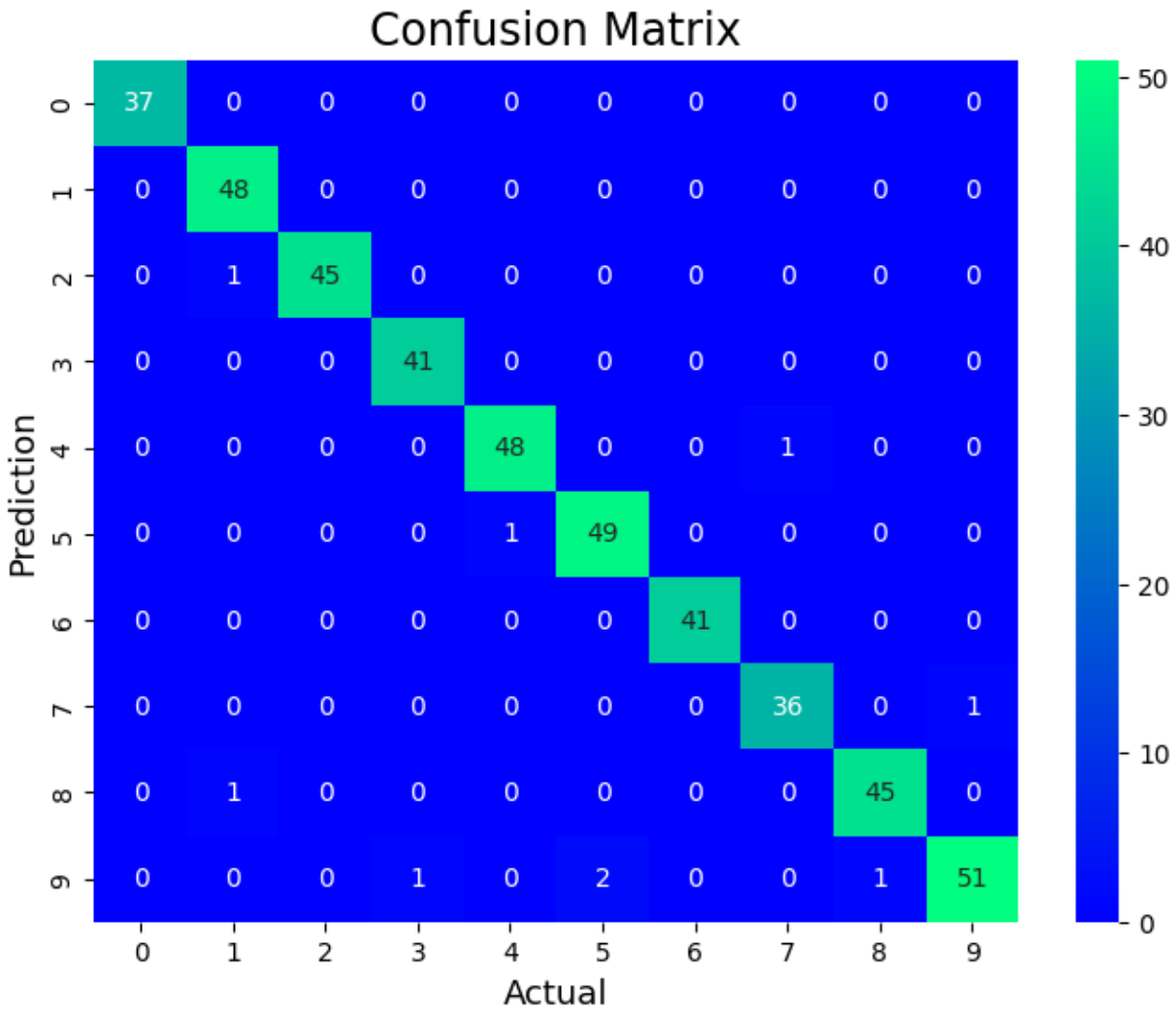
Aim: To analyse the performance of a multi class confusion matrix by using choosen database with python code .

Program:

```
from sklearn.datasets import load_digits

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import confusion_matrix
```

```
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
X, y = load_digits(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=23)
clf = RandomForestClassifier(random_state=23)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='g', cmap="winter")
plt.ylabel('Prediction', fontsize=13)
plt.xlabel('Actual', fontsize=13)
plt.title('Confusion Matrix', fontsize=17)
plt.show()
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, average='weighted')
recall = recall_score(y_test, y_pred, average='weighted')
f1 = f1_score(y_test, y_pred, average='weighted')
print("Accuracy  :", accuracy)
print("Precision :", precision)
print("Recall    :", recall)
print("F1 Score  :", f1)
```

OUTPUT:



Accuracy : 0.98

Precision : 0.9801896286719816

Recall : 0.98

F1 Score : 0.9799237684030243

Experiment 4:

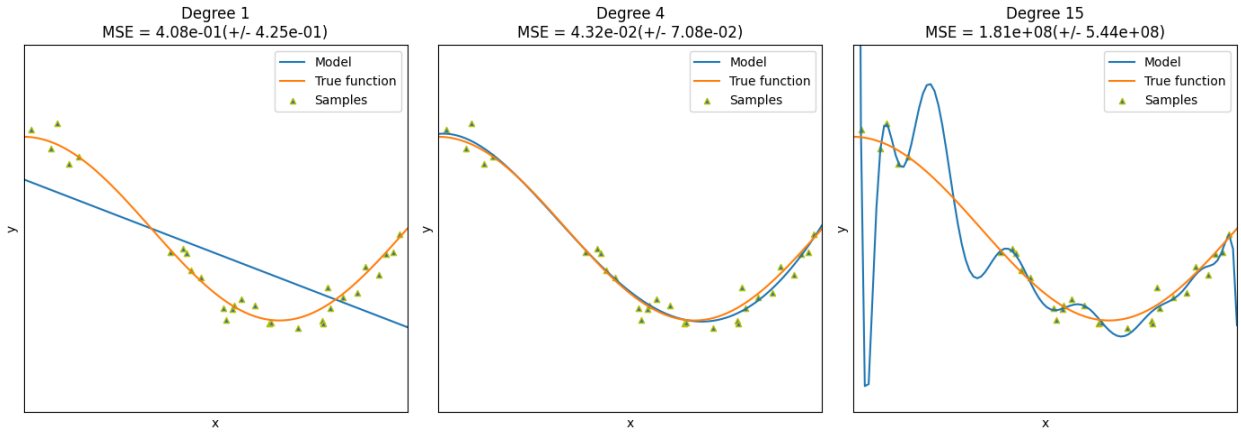
Aim: To analyse the performance of a over fitting by using choosen database with python code .

Program:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import cross_val_score
def true_fun(X):
    return np.cos(1.5 * np.pi * X)
np.random.seed(0)
n_samples = 30
degrees = [1, 4, 15]
X = np.sort(np.random.rand(n_samples))
y = true_fun(X) + np.random.randn(n_samples) * 0.1
plt.figure(figsize=(14, 5))
for i in range(len(degrees)):
    ax = plt.subplot(1, len(degrees), i + 1)
    plt.setp(ax, xticks=(), yticks=())
    polynomial_features = PolynomialFeatures(degree=degrees[i], include_bias=False)
    linear_regression = LinearRegression()
    pipeline = Pipeline(
        [
            ("polynomial_features", polynomial_features),
            ("linear_regression", linear_regression),
        ]
    )
    pipeline.fit(X[:, np.newaxis], y)
```

```
    scores = cross_val_score(
        pipeline, X[:, np.newaxis], y, scoring="neg_mean_squared_error", cv=10
    )
    X_test = np.linspace(0, 1, 100)
    plt.plot(X_test, pipeline.predict(X_test[:, np.newaxis]), label="Model")
    plt.plot(X_test, true_fun(X_test), label="True function")
    plt.scatter(X, y, edgecolor="y", s=20, marker="^", label="Samples")
    plt.xlabel("x")
    plt.ylabel("y")
    plt.xlim((0, 1))
    plt.ylim((-2, 2))
    plt.legend(loc="best")
    plt.title(
        "Degree {}\nMSE = {:.2e}(+/- {:.2e})".format(
            degrees[i], -scores.mean(), scores.std()
        )
    )

plt.tight_layout()
plt.show()
```

OUTPUT:



Experiment 5:

Aim: To demonstrate the performance of a linear regression by using choosen database with python code .

Program:

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error

iris = load_iris()

data = pd.DataFrame(data=iris.data, columns=iris.feature_names)

data['species'] = iris.target

print(data.head())

X = data[['sepal length (cm)']]

y = data['sepal width (cm)']


X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

model = LinearRegression()
```

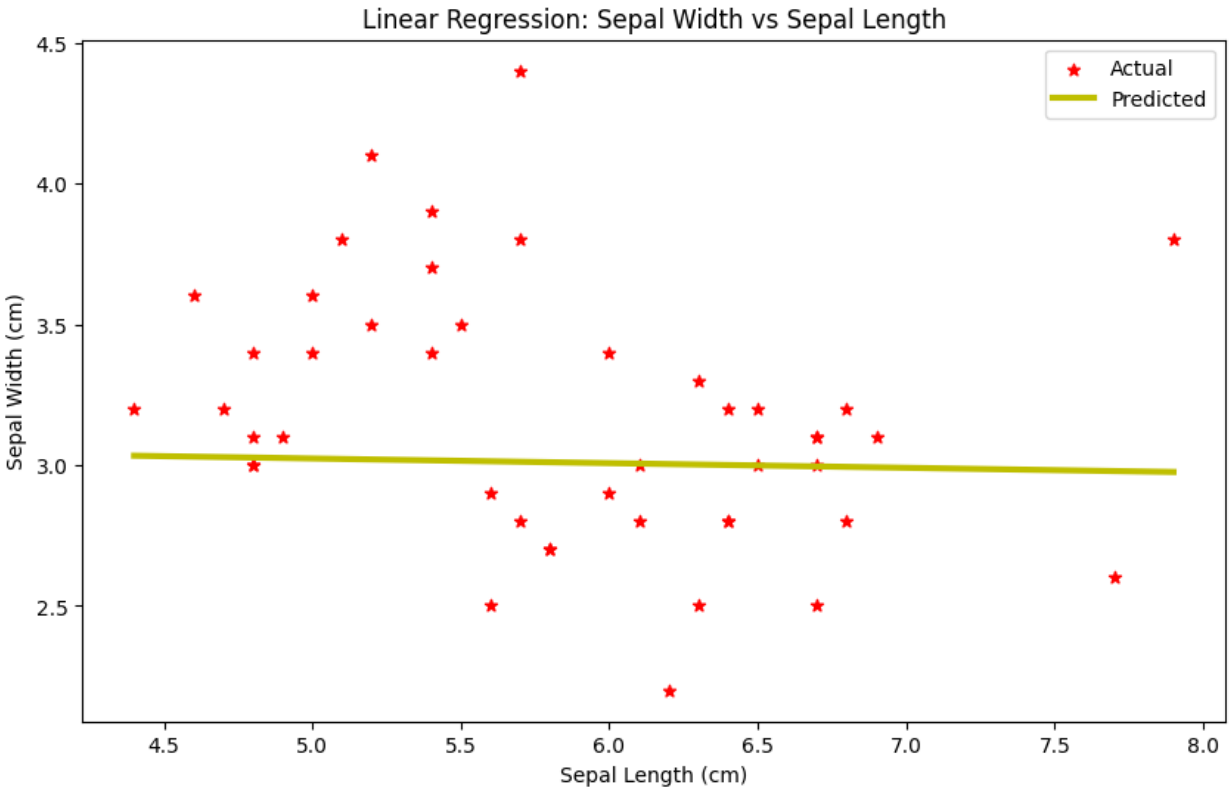
```
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)

print(f'Mean Squared Error on test set: {mse:.2f}')

plt.figure(figsize=(10, 6))

plt.scatter(X_test, y_test, color='r', marker='*', label='Actual')

# Sort X_test and corresponding y_pred for a clean line plot

sort_indices = X_test['sepal length (cm)'].argsort()

plt.plot(X_test.iloc[sort_indices], y_pred[sort_indices], color='y', linewidth=3,
label='Predicted')

plt.xlabel('Sepal Length (cm)')

plt.ylabel('Sepal Width (cm)')

plt.title('Linear Regression: Sepal Width vs Sepal Length')

plt.legend()

plt.show()

new_sample = pd.DataFrame([[5]], columns=['sepal length (cm)'])

predicted_width = model.predict(new_sample)

print(f'The predicted sepal width for sepal length {new_sample.values.tolist()} is
{predicted_width[0]:.2f} cm')
```

OUTPUT:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	\
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	
4	5.0	3.6	1.4	0.2	

	species
0	0
1	0
2	0
3	0
4	0

Mean Squared Error on test set: 0.23



The predicted sepal width for sepal length [[5]] is 3.02 cm